

Overview of materials for Acrylonitrile Butadiene Styrene (ABS), Molded

Categories: [Polymer](#); [Thermoplastic](#); [ABS Polymer](#); [Acrylonitrile Butadiene Styrene \(ABS\), Molded](#)

Material Notes: This property data is a summary of similar materials in the MatWeb database for the category "Acrylonitrile Butadiene Styrene (ABS), Molded". Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

Vendors: [Click here to view all available suppliers for this material.](#)

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Physical Properties	Metric	English	Comments
Density	0.882 - 3.50 g/cc	0.0319 - 0.126 lb/in ³	Average value: 1.07 g/cc Grade Count:774
Water Absorption	0.0250 - 2.30 %	0.0250 - 2.30 %	Average value: 0.383 % Grade Count:85
Moisture Absorption at Equilibrium	0.100 - 0.300 %	0.100 - 0.300 %	Average value: 0.232 % Grade Count:26
Water Absorption at Saturation	0.00950 - 1.03 %	0.00950 - 1.03 %	Average value: 0.561 % Grade Count:14
Maximum Moisture Content	0.0100 - 0.150	0.0100 - 0.150	Average value: 0.0459 Grade Count:71
Linear Mold Shrinkage	0.000 - 0.0290 cm/cm	0.000 - 0.0290 in/in	Average value: 0.00533 cm/cm Grade Count:581
Linear Mold Shrinkage, Transverse	0.00200 - 0.00900 cm/cm	0.00200 - 0.00900 in/in	Average value: 0.00522 cm/cm Grade Count:52
Melt Flow	0.0800 - 125 g/10 min	0.0800 - 125 g/10 min	Average value: 16.5 g/10 min Grade Count:805

Mechanical Properties	Metric	English	Comments
Hardness, Rockwell M	53.0 - 92.0	53.0 - 92.0	Average value: 71.8 Grade Count:4
Hardness, Rockwell R	13.0 - 122	13.0 - 122	Average value: 108 Grade Count:431
Hardness, Shore D	79.0 - 103	79.0 - 103	Average value: 87.0 Grade Count:3
Ball Indentation Hardness	70.0 - 120 MPa	10200 - 17400 psi	Average value: 101 MPa Grade Count:68
Tensile Strength, Ultimate	2.60 - 73.1 MPa	377 - 10600 psi	Average value: 41.0 MPa Grade Count:346
 Tensile Strength, Yield	20.0 - 43.0 MPa @ Temperature 60.0 - 90.0 °C	2900 - 6240 psi @ Temperature 140 - 194 °F	Average value: 31.8 MPa Grade Count:2
Tensile Strength, Yield	2.00 - 77.0 MPa	290 - 11200 psi	Average value: 44.8 MPa Grade Count:624
Elongation at Break	1.40 - 110 %	1.40 - 110 %	Average value: 28.5 % Grade Count:493

Elongation at Yield	1.70 - 40.0 %	1.70 - 40.0 %	Average value: 3.66 % Grade Count:189
Modulus of Elasticity	0.778 - 21.2 GPa	113 - 3080 ksi	Average value: 2.35 GPa Grade Count:356
Flexural Yield Strength	0.379 - 655 MPa	55.0 - 95000 psi	Average value: 70.6 MPa Grade Count:626
Flexural Modulus	0.0241 - 6.89 GPa	3.50 - 1000 ksi	Average value: 2.33 GPa Grade Count:699
	1.50 - 4.00 GPa @Temperature 60.0 - 90.0 °C	218 - 580 ksi @Temperature 140 - 194 °F	Average value: 2.84 GPa Grade Count:2
Poissons Ratio	0.360 - 0.380	0.360 - 0.380	Average value: 0.364 Grade Count:5
Izod Impact, Notched	0.100 - 7.85 J/cm	0.187 - 14.7 ft-lb/in	Average value: 2.41 J/cm Grade Count:516
	0.350 - 3.00 J/cm @Temperature -50.0 - 0.000 °C	0.656 - 5.62 ft-lb/in @Temperature -58.0 - 32.0 °F	Average value: 1.00 J/cm Grade Count:23
	0.290 - 2.35 J/cm @Temperature -40.0 - 0.000 °C	0.543 - 4.40 ft-lb/in @Temperature -40.0 - 32.0 °F	Average value: 1.00 J/cm Grade Count:30
	0.290 - 2.35 J/cm @Thickness 3.17 - 12.7 mm	0.543 - 4.40 ft-lb/in @Thickness 0.125 - 0.500 in	Average value: 1.00 J/cm Grade Count:30
Izod Impact, Unnotched	1.50 J/cm - NB	2.81 ft-lb/in - NB	Average value: 8.79 J/cm Grade Count:27
	0.600 - 6.47239 J/cm @Temperature -30.0 - 0.125 °C	1.12 - 12.1254 ft-lb/in @Temperature -22.0 - 32.2 °F	Average value: 1.24 J/cm Grade Count:4
	0.392266 - 4.8051 J/cm @Temperature -30.0 - -20.0 °C	0.734875 - 9.0019 ft-lb/in @Temperature -22.0 - -4.00 °F	Average value: 1.24 J/cm Grade Count:34
	0.392266 - 4.8051 J/cm @Thickness 3.20 - 6.40 mm	0.734875 - 9.0019 ft-lb/in @Thickness 0.126 - 0.252 in	Average value: 1.24 J/cm Grade Count:34
Izod Impact, Notched (ISO)	1.00 - 45.0 kJ/m²	0.476 - 21.4 ft-lb/in²	Average value: 18.5 kJ/m² Grade Count:184
	6.00 - 22.0 kJ/m² @Temperature -40.0 - -20.0 °C	2.86 - 10.5 ft-lb/in² @Temperature -40.0 - -4.00 °F	Average value: 9.13 kJ/m² Grade Count:61
	6.00 - 6.00 kJ/m² @Temperature -40.0 - -40.0 °C	2.86 - 2.86 ft-lb/in² @Temperature -40.0 - -40.0 °F	Average value: 9.13 kJ/m² Grade Count:1
	6.00 - 6.00 kJ/m² @Thickness 4.00 - 4.00 mm	2.86 - 2.86 ft-lb/in² @Thickness 0.157 - 0.157 in	Average value: 9.13 kJ/m² Grade Count:1
Izod Impact, Unnotched (ISO)	39.2 kJ/m² - NB	18.7 ft-lb/in² - NB	Average value: 77.2 kJ/m² Grade Count:27
	60.0 - 70.0 kJ/m² @Temperature -30.0 - -30.0 °C	28.6 - 33.3 ft-lb/in² @Temperature -22.0 - -22.0 °F	Average value: 68.0 kJ/m² Grade Count:5
Charpy Impact Unnotched	1.00 J/cm² - NB	4.76 ft-lb/in² - NB	Average value: 11.4 J/cm² Grade Count:145
	0.300 J/cm² - NB @Temperature -40.0 - -20.0 °C	1.43 ft-lb/in² - NB @Temperature -40.0 - -4.00 °F	Average value: 7.48 J/cm² Grade Count:96
Charpy Impact, Notched	0.100 - 14.0 J/cm²	0.476 - 66.6 ft-lb/in²	Average value: 1.90 J/cm² Grade Count:292
	0.200 - 2.40 J/cm² @Temperature -40.0 - -20.0 °C	0.952 - 11.4 ft-lb/in² @Temperature -40.0 - -4.00 °F	Average value: 0.877 J/cm² Grade Count:103
Gardner Impact	1.80 - 22.6 J	1.33 - 16.7 ft-lb	Average value: 15.9 J Grade Count:7
Dart Drop, Total Energy	2.37 - 46.3 J	1.75 - 34.1 ft-lb	Average value: 28.0 J Grade Count:5
	32.8 - 32.8 J @Temperature -18.0 - -18.0 °C	24.2 - 24.2 ft-lb @Temperature -0.400 - -0.400 °F	Average value: 35.8 J Grade Count:1
	37.3 - 37.3 J @Temperature -18.0 - -18.0 °C	27.5 - 27.5 ft-lb @Temperature -0.400 - -0.400 °F	Average value: 35.8 J Grade Count:1
	37.3 - 37.3 J @Thickness 3.20 - 3.20 mm	27.5 - 27.5 ft-lb @Thickness 0.126 - 0.126 in	Average value: 35.8 J Grade Count:1

Falling Dart Impact	2.82 - 37.6 J	2.08 - 27.7 ft-lb	Average value: 25.3 J Grade Count:9
Instrumented Impact Total Energy	5.40 - 54.0 J	3.98 - 39.8 ft-lb	Average value: 39.6 J Grade Count:21
Tensile Creep Modulus, 1 hour	2200 - 2800 MPa	319000 - 406000 psi	Average value: 2480 MPa Grade Count:4
Tensile Creep Modulus, 1000 hours	1250 - 1900 MPa	181000 - 276000 psi	Average value: 1640 MPa Grade Count:4
Electrical Properties	Metric	English	Comments
Electrical Resistivity	1500 - 1.00e+18 ohm-cm	1500 - 1.00e+18 ohm-cm	Average value: 1.87e+16 ohm-cm Grade Count:127
Surface Resistance	1000 - 2.00e+17 ohm	1000 - 2.00e+17 ohm	Average value: 6.94e+15 ohm Grade Count:120
Static Decay	0.250 - 3.00 sec	0.250 - 3.00 sec	Average value: 1.24 sec Grade Count:7
Dielectric Constant	2.70 - 3.80	2.70 - 3.80	Average value: 2.98 Grade Count:54
Dielectric Strength	15.7 - 53.0 kV/mm	400 - 1350 kV/in	Average value: 32.0 kV/mm Grade Count:80
Dissipation Factor	0.00400 - 0.0900	0.00400 - 0.0900	Average value: 0.00949 Grade Count:52
Arc Resistance	0.000 - 180 sec	0.000 - 180 sec	Average value: 78.8 sec Grade Count:28
Comparative Tracking Index	92.0 - 600 V	92.0 - 600 V	Average value: 561 V Grade Count:93
Hot Wire Ignition, HWI	7.00 - 120 sec	7.00 - 120 sec	Average value: 26.9 sec Grade Count:33
High Amp Arc Ignition, HAI	30.0 - 200 arcs	30.0 - 200 arcs	Average value: 123 arcs Grade Count:33
High Voltage Arc-Tracking Rate, HVTR	0.000 - 150 mm/min	0.000 - 5.91 in/min	Average value: 34.4 mm/min Grade Count:31
Thermal Properties	Metric	English	Comments
CTE, linear	7.90 - 139 µm/m-°C	4.39 - 77.2 µin/in-°F	Average value: 83.4 µm/m-°C Grade Count:189
CTE, linear, Transverse to Flow	81.0 - 100 µm/m-°C	45.0 - 55.6 µin/in-°F	Average value: 91.4 µm/m-°C Grade Count:15
Specific Heat Capacity	1.60 - 2.13 J/g-°C	0.382 - 0.509 BTU/lb-°F	Average value: 1.99 J/g-°C Grade Count:7
Thermal Conductivity	0.128 - 0.187 W/m-K	0.888 - 1.30 BTU-in/hr-ft²-°F	Average value: 0.162 W/m-K Grade Count:19
 @Temperature 30.0 - 260 °C	0.250 - 0.250 W/m-K	1.74 - 1.74 BTU-in/hr-ft²-°F @Temperature 86.0 - 500 °F	Average value: 0.250 W/m-K Grade Count:1
Maximum Service Temperature, Air	50.0 - 109 °C	122 - 228 °F	Average value: 74.9 °C Grade Count:24
Hot Ball Pressure Test	75.0 - 105 °C	167 - 221 °F	Average value: 88.0 °C Grade Count:27

Deflection Temperature at 0.46 MPa (66 psi)	56.0 - 165 °C	133 - 329 °F	Average value: 95.3 °C Grade Count:256
Deflection Temperature at 1.8 MPa (264 psi)	63.9 - 220 °C	147 - 428 °F	Average value: 89.0 °C Grade Count:702
Vicat Softening Point	45.0 - 160 °C	113 - 320 °F	Average value: 99.5 °C Grade Count:571
Heat Distortion Temperature	76.4 - 87.8 °C	170 - 190 °F	Average value: 84.2 °C Grade Count:6
Glass Transition Temp, Tg	105 - 109 °C	221 - 228 °F	Average value: 107 °C Grade Count:9
UL RTI, Electrical	50.0 - 120 °C	122 - 248 °F	Average value: 73.4 °C Grade Count:94
UL RTI, Mechanical with Impact	50.0 - 105 °C	122 - 221 °F	Average value: 71.1 °C Grade Count:85
UL RTI, Mechanical without Impact	50.0 - 120 °C	122 - 248 °F	Average value: 73.2 °C Grade Count:84
Flammability, UL94	HB - 5VA	HB - 5VA	Average value: 1.50 Grade Count:511
Oxygen Index	19.0 - 30.0 %	19.0 - 30.0 %	Average value: 21.6 % Grade Count:5
Glow Wire Test	600 - 960 °C	1110 - 1760 °F	Average value: 673 °C Grade Count:49
Optical Properties	Metric	English	Comments
Haze	2.00 - 4.00 %	2.00 - 4.00 %	Average value: 2.74 % Grade Count:7
Gloss	30.0 - 98.0 %	30.0 - 98.0 %	Average value: 84.0 % Grade Count:57
Yellow Index	-1.40 - 24.3 %	-1.40 - 24.3 %	Average value: 14.2 % Grade Count:5
Transmission, Visible	0.000 - 90.0 %	0.000 - 90.0 %	Average value: 68.8 % Grade Count:51
Processing Properties	Metric	English	Comments
Processing Temperature	170 - 270 °C	338 - 518 °F	Average value: 212 °C Grade Count:87
Nozzle Temperature	180 - 310 °C	356 - 590 °F	Average value: 238 °C Grade Count:172
Adapter Temperature	200 - 300 °C	392 - 572 °F	Average value: 277 °C Grade Count:31
Die Temperature	200 - 295 °C	392 - 563 °F	Average value: 267 °C Grade Count:31
Melt Temperature	149 - 323 °C	300 - 613 °F	Average value: 234 °C Grade Count:315
Mold Temperature	10.0 - 120 °C	50.0 - 248 °F	Average value: 62.3 °C Grade Count:377
Injection Velocity	200 - 240 mm/sec	7.87 - 9.45 in/sec	Average value: 232 mm/sec Grade Count:30
Roll Temperature	60.0 - 150 °C	140 - 302 °F	Average value: 127 °C Grade Count:29

Drying Temperature	60.0 - 120 °C	140 - 248 °F	Average value: 84.2 °C Grade Count:381
Moisture Content	0.0100 - 0.300 %	0.0100 - 0.300 %	Average value: 0.0668 % Grade Count:115
Dew Point	-29.0 - -17.8 °C	-20.2 - 0.000 °F	Average value: -19.6 °C Grade Count:6
Injection Pressure	4.14 - 130 MPa	600 - 18900 psi	Average value: 56.9 MPa Grade Count:35
Vent Depth	0.00254 - 0.0510 cm	0.00100 - 0.0201 in	Average value: 0.0123 cm Grade Count:10

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.